UNIT 1

Introduction

Sessions 1 and 2

Before you write a single line of code you need two things: the names of the parts in front of you, and a clear idea of how electricity moves. This unit has no programming at all - and yet by the end of it you will have made a light glow.

By the end of this unit you will be able to:

- Name every part in your kit and say what it is for.
- Explain the Sense - Think - Act loop with your own example.
- Draw a circuit and explain why it must be a complete loop.
- Use a breadboard correctly and light an LED with no code.

SESSION 1

one class

What Is a Robot, and What Is in Your Box

Today you will

- Learn what really makes a machine a robot.
- Open your kit and learn the name of every part.
- Agree the seven rules of the robotics lab.

When somebody says the word *robot*, most people picture a metal human with glowing eyes. Real robots almost never look like that. A robot is any machine that can **sense** something about the world, **decide** what that means, and then **act** on the decision.

By that definition there are robots all around you already. The automatic hand dryer in a mall senses your hands, decides they are there, and blows air. A washing machine senses that the drum is unbalanced, decides to slow down, and does it. Neither of them looks like a robot. Both of them behave like one.

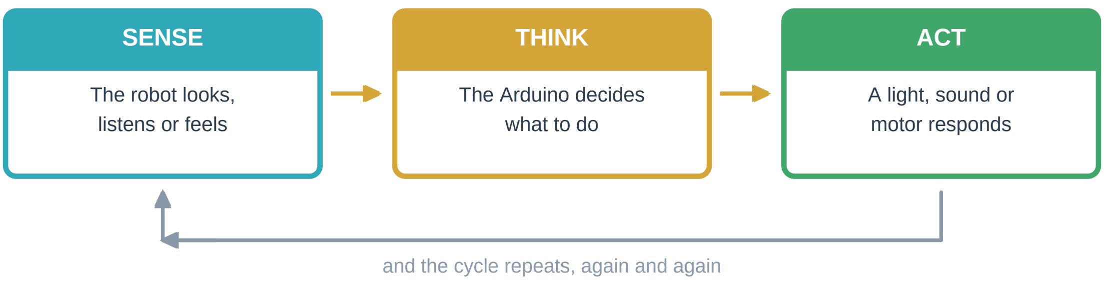


Figure 1.1 - Every robot in the world follows this same three-step loop

Every single thing you build this year follows this loop. In Unit 3 your machine will only act - it will have no senses at all. In Unit 4 it gets its first sense. In Unit 5 it senses, thinks and moves a real object. Keep this diagram in your head; you will come back to it many times.

Try This

Take one minute and look around the classroom. Find three machines that sense something. For each, write down what it senses, what it decides, and what it does. A fan regulator, a mobile phone screen and a fire alarm are good places to start.

1.1 Open the Box

Now open your kit and lay everything out on the desk. Do not plug anything in yet. Your only job in this session is to learn the names. Half of robotics is being able to say exactly what you are holding - it is impossible to ask for help with 'that small blue thing'.

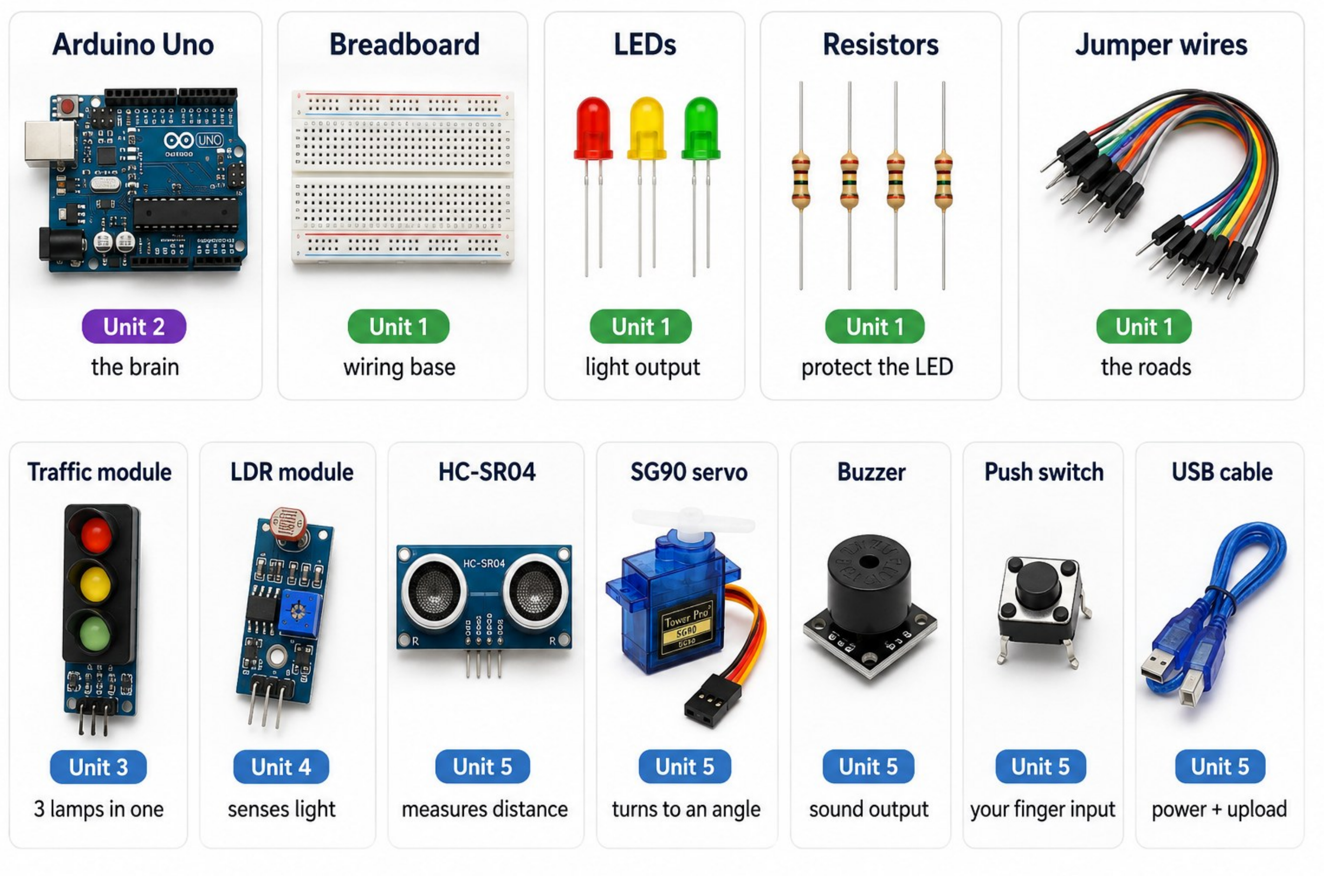


Figure 1.1 - Your kit. Learn these twelve names before the next class.

Part	What it does	First used in
Arduino Uno board	The brain. It remembers your program and follows it forever, even after the power has been off.	Session 3
USB cable	Carries your program from the laptop into the board, and powers the board while you test.	Session 3
Breadboard	A plastic block full of holes that joins parts together without soldering.	Session 2
Jumper wires	The roads between parts. They come as male-to-male, male-to-female and female-to-female.	Session 2
LEDs	Tiny lamps. They glow only one way round.	Session 2
Resistors	Slow the current down so the LED is not destroyed.	Session 2
Traffic light module	Red, yellow and green lamps already built onto one small board.	Session 7
LDR sensor module	Feels how bright the room is and turns it into a number.	Session 9

Part	What it does	First used in
HC-SR04 ultrasonic sensor	Measures how far away an object is, without touching it.	Session 12
SG90 servo motor	Turns to an exact angle and holds it. This will be your barrier arm.	Session 12
Buzzer	Makes sound, so your machine can warn people.	Session 13
Push switch	Lets a human tell the machine something.	Challenge work

1.2 Rules of the Robotics Lab

These seven rules are not decoration. Each one exists because somebody, somewhere, broke a board by ignoring it. Read them once now and follow them for the rest of the year.

1. Always unplug the USB cable before you change any wire.
2. Check the long leg of every LED and the + side of every module before power goes on.
3. Never run a wire straight from 5V to GND. That is a short circuit and it can kill the board.
4. Hold the Arduino by its edges. Do not touch the metal pins with wet hands.
5. One person wires, one person checks, one person writes the notebook. Then swap roles.
6. If anything smells hot or feels hot, unplug it at once and call your trainer.
7. Count every component back into the kit box before you leave.

> **Watch Out**
> A circuit that does not work is completely normal. It is not failure - it is step one of **debugging**. Professional engineers spend more time finding faults than building. Every fault you track down teaches you something a working circuit never will.

New word	What it means
Robot	A machine that senses, decides and acts.
Component	Any single part in your kit.
Debugging	Hunting down the reason something does not work.

SESSION 2 one class

Electricity, Circuits and the Breadboard

Today you will

- Understand voltage, current and resistance without any maths.
- Learn why a circuit must be a complete loop.
- Light an LED using the breadboard and no code at all.

Electricity is invisible, which makes it hard to imagine. So engineers borrow a picture from something you can see: water in a pipe. The battery is the pump, the wires are the pipes, and the LED is a small fountain at the far end. Once you hold that picture, three words start to make sense.

Word	In the water picture	In electricity
Voltage, measured in volts (V)	How hard the pump pushes	How hard the battery pushes electricity along. Your Arduino pushes at 5V.
Current, measured in amps (A)	How much water flows every second	How much electricity flows every second. Too much current burns parts out.

Word	In the water picture	In electricity
Resistance, measured in ohms	A narrow section of pipe	A part that makes it harder for electricity to flow. That is a resistor's whole job.

Now the single most important rule in this book: **electricity only flows in a complete loop**. It has to leave the battery, travel through every part, and arrive back at the battery. Break the loop anywhere and everything stops instantly. That is not a fault - it is exactly how a switch works. A switch is simply a gap you can open and close on purpose.

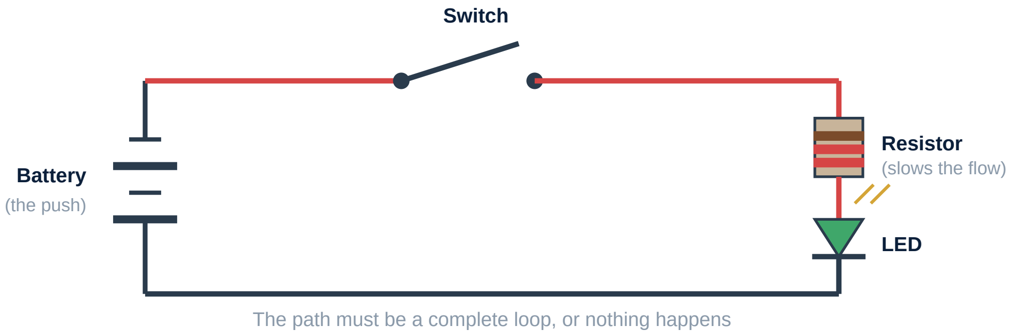
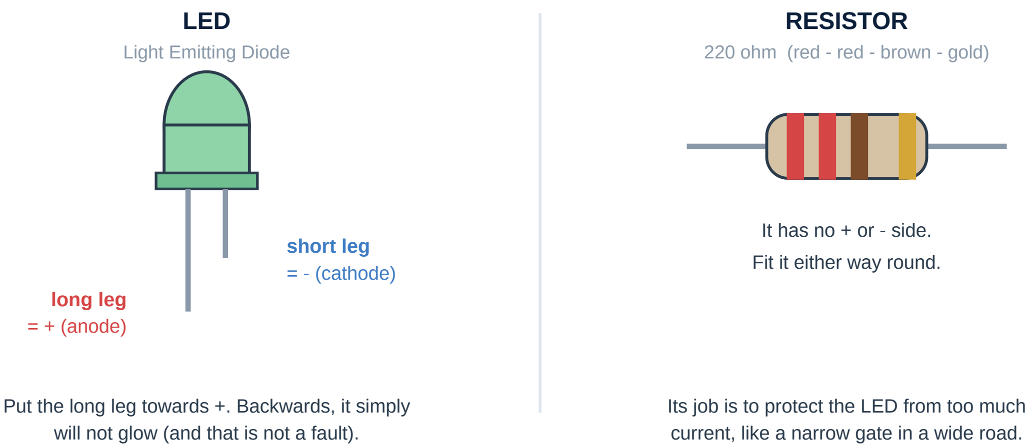


Figure 1.2 - A simple circuit: battery, switch, resistor and LED

2.1 The Two Parts You Will Use Most

An LED and a resistor almost always travel together. The LED gives you light. The resistor keeps the LED alive. Connect an LED to 5V with no resistor and it receives far more current than it can survive - it will glow very brightly for about a second, and then never again.



Watch Out

An LED has a long leg and a short leg. Long leg towards +, short leg towards -. If your LED does not glow, turn it around before you check anything else. Nine times out of ten that is the entire problem.

2.2 Inside the Breadboard

A breadboard looks like a boring block of holes. It is actually full of hidden metal strips. Two holes joined by a hidden strip behave exactly as if you had twisted those two wires together. Knowing which holes are secretly joined is the most useful thing in this whole unit - and the commonest reason a beginner's circuit does nothing at all.

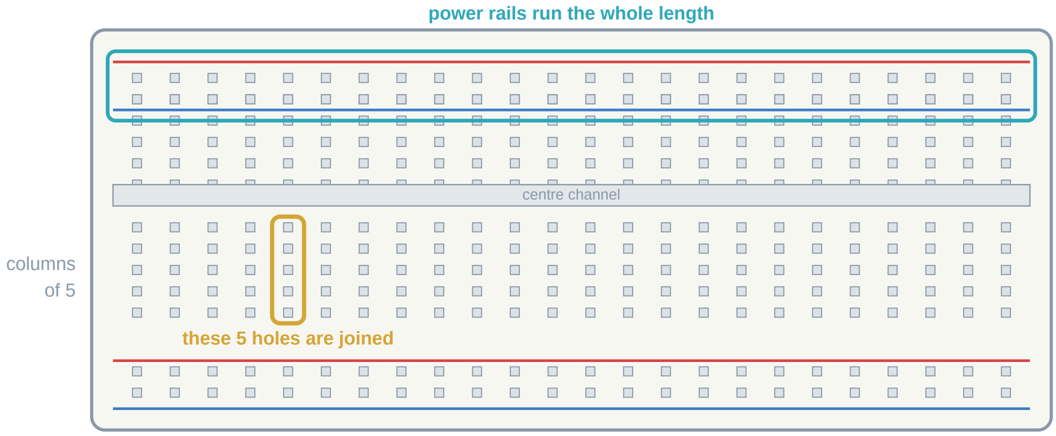


Figure 1.3 - Inside a breadboard: which holes are secretly connected

- The **long side rails**, marked + and -, run the entire length of the board. Use them for power and ground so that every part can reach them.
- In the middle, holes are joined in **columns of five**, up and down only - never across.
- The **centre channel** splits the board into two halves. Nothing crosses it.
- So: two legs in the same column of five are connected. Two legs in different columns are not. That is the whole secret.

HANDS ON

Activity - Light an LED With No Code At All

Here you will not use the Arduino as a computer. You will use it only as a battery, to prove to yourself that you understand a circuit.

1. Push an LED into the breadboard so its two legs sit in **different** columns.
2. Push one leg of a 220 ohm resistor into the same column as the LED's **long** leg.
3. Run a jumper wire from the resistor's free leg to the **5V** pin on the Arduino.
4. Run another jumper wire from the LED's **short** leg column to any **GND** pin.
5. Now plug in the USB cable. The LED should glow steadily.
6. Pull out the GND wire. The light dies - you have just broken the loop, exactly like a switch.

If it does not glow, check in this order

1. Is the LED the right way round? 2. Are both legs in different columns? 3. Is the resistor really in the same column as the long leg? 4. Is the USB firmly plugged in at both ends?

Did You Know

LED stands for Light Emitting Diode. A diode is a one-way gate for electricity. That is the real reason an LED refuses to work backwards - it is not broken, it is doing its job perfectly.

New word	What it means
Voltage	How hard electricity is pushed. Measured in volts.
Current	How much electricity flows each second. Measured in amps.
Resistor	A part that limits current. Measured in ohms.
Ground (GND)	The return path. Every circuit needs one.